

A useful theorem for designing controlled unitary gates

When no X in between: $\overbrace{R_z(\mu) R_y\left(\frac{\nu}{2}\right)}^A \overbrace{R_y\left(-\frac{\nu}{2}\right) R_z\left(\frac{-\xi-\mu}{2}\right)}^B \overbrace{R_z\left(\frac{\xi-\mu}{2}\right)}^C = \underline{\underline{I}}$

Handwritten notes:
 $R_y(0) = I$
 $R_z(0) = I$
 $R_z(-\mu)$

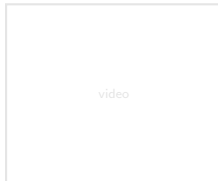
When X is in between: $\overbrace{R_z(\mu) R_y\left(\frac{\nu}{2}\right)}^A X \overbrace{R_y\left(-\frac{\nu}{2}\right) R_z\left(\frac{-\xi-\mu}{2}\right)}^B X \overbrace{R_z\left(\frac{\xi-\mu}{2}\right)}^C = \underline{R_z(\mu) R_y(\nu) R_z(\xi)} = U$

Handwritten notes:
 $XYX = -Y$
 $R_y(\nu)$
 $R_y\left(\frac{\nu}{2}\right)$
 $XX = I$
 $R_z\left(\frac{\xi+\mu}{2}\right)$
 $R_z\left(\frac{\xi}{2}\right)$

$$X R_y\left(-\frac{\nu}{2}\right) X = X \left[\cos \frac{\nu}{4} I + i \sin \frac{\nu}{4} Y \right] X$$

$$= \cos \frac{\nu}{4} I - i \sin \frac{\nu}{4} Y = R_y\left(\frac{\nu}{2}\right)$$

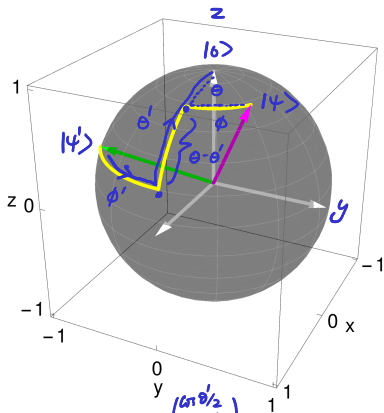
$$XZX = -Z$$



How to go from one pure state to another in 3 rotations

$$R_{\eta}^{-1}(\theta) = R_{\eta}^{\dagger}(\theta) = R_{-\eta}(\theta) = R_{\eta}(-\theta)$$

$$R_{\eta}(\theta_1) R_{\eta}(\theta_2) = R_{\eta}(\theta_2) R_{\eta}(\theta_1) = R_{\eta}(\theta_1 + \theta_2)$$



$$\eta_1 \neq \eta_2$$

$$R_{\eta_2}(\theta_2) R_{\eta_1}(\theta_1) \neq R_{\eta_1}(\theta_1) R_{\eta_2}(\theta_2)$$

$$XY = iZ$$

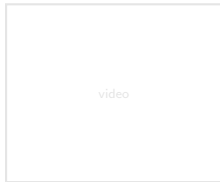
$$YX = -iZ$$

$$\mathcal{B} = \{|0\rangle, |1\rangle\}$$

$$\begin{pmatrix} \cos \theta/2 \\ \sin \theta/2 \end{pmatrix}$$

$$\begin{pmatrix} \cos \theta'/2 \\ \sin \theta'/2 \end{pmatrix}$$

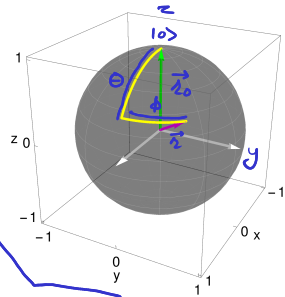
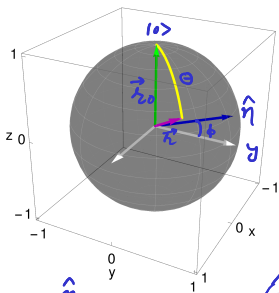
$$\underbrace{\begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix}}_{|\psi\rangle} = \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix}}_{\substack{R_z(\phi) \\ \text{PS}}} \underbrace{\begin{pmatrix} \cos \frac{\theta - \theta'}{2} & -\sin \frac{\theta - \theta'}{2} \\ \sin \frac{\theta - \theta'}{2} & \cos \frac{\theta - \theta'}{2} \end{pmatrix}}_{\substack{R_y(\theta - \theta') \\ \text{RS}}} \underbrace{\begin{pmatrix} 1 & 0 \\ 0 & e^{-i\phi'} \end{pmatrix}}_{\substack{R_z(-\phi') \\ \text{Phase shifter}}} \underbrace{\begin{pmatrix} \cos \frac{\theta'}{2} \\ \sin \frac{\theta'}{2} e^{i\phi'} \end{pmatrix}}_{|\psi'\rangle}$$



How to prepare a general pure state

$$|4\rangle = \cos \frac{\theta}{2} |0\rangle + \sin \frac{\theta}{2} e^{i\phi} |1\rangle$$

$$\begin{aligned} |4\rangle &= |0\rangle \\ \rho &= |4\rangle\langle 4| \\ &= \frac{I + \vec{r}_0 \cdot \vec{\sigma}}{2} \\ &= \frac{I + Z}{2} \\ \vec{r}_0 &= (0, 0, 1) \end{aligned}$$



$$\begin{aligned} \rho &= |4\rangle\langle 4| \\ &= \frac{I + \vec{r} \cdot \vec{\sigma}}{2} \\ \vec{r} &= (\sin \theta \cos \phi, \sin \theta \sin \phi, \cos \theta) \end{aligned}$$



$$\vec{r} = \hat{z} \times \vec{r} = \sin \theta (-\sin \phi, \cos \phi, 0)$$

$$R_{\eta}(\theta) = \cos \frac{\theta}{2} I - i \sin \frac{\theta}{2} \hat{\eta} \cdot \vec{\sigma}$$

$$|0\rangle \xrightarrow{R_1(\theta)} \xrightarrow{R_2(\phi)} |4\rangle$$

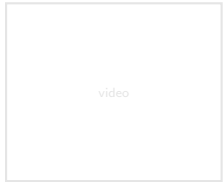
$$\begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix} = \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} e^{-i\phi} \\ \sin \frac{\theta}{2} e^{i\phi} & \cos \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\mathcal{B} = \{|0\rangle, |1\rangle\}$$

$$|0\rangle \xrightarrow{R_{\eta}(\theta)} |4\rangle$$

$$\begin{pmatrix} \cos \frac{\theta}{2} \\ \sin \frac{\theta}{2} e^{i\phi} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\phi} \end{pmatrix} \begin{pmatrix} \cos \frac{\theta}{2} & -\sin \frac{\theta}{2} \\ \sin \frac{\theta}{2} & \cos \frac{\theta}{2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

Beam splitter
Phase shifter
 $R_y(\theta)|0\rangle$
 $R_2(\phi)$
 $R_y(\theta)$
 $|0\rangle$

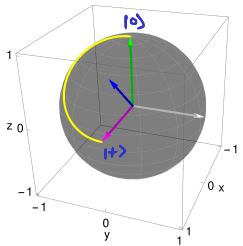
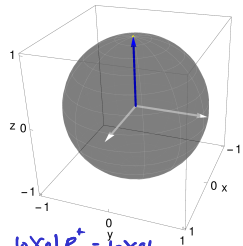
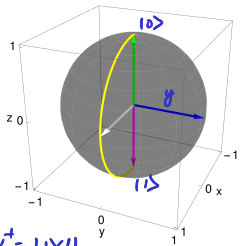
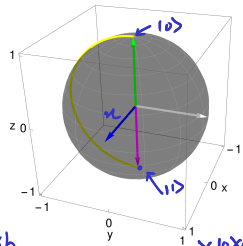


Pauli, Hadamard, and Phase gates

$\vec{z}_0 = (0,0,1)$ $|4_0\rangle = |0\rangle$

$H|0\rangle = |+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$
 $H|1\rangle = |-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)$

$t = \pi$



$X|+\rangle = +|1\rangle$
 $X|-\rangle = -|1\rangle$

$t = \pi$

Bit flip
NOT gate

$$X = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

$X|0\rangle = |1\rangle$
 $X|1\rangle = |0\rangle$

$Y|0\rangle = i|1\rangle$
 $Y|1\rangle = -i|0\rangle$

$$Y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$

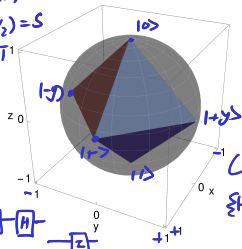
$B = \{|0\rangle, |1\rangle\}$

$R_z|0\rangle = |0\rangle$
 $R_z|1\rangle = |1\rangle$

$$R_z(t) = \begin{pmatrix} 1 & 0 \\ 0 & e^{it} \end{pmatrix}$$

$R_z(\pi) = Z$
 $R_z(\pi/2) = S$

$R_z(\pi/4) = T$
 $\pi/8$ gate



$H = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix} = \frac{1}{\sqrt{2}}(1,0,1) \cdot \vec{\sigma}$
 $H|0\rangle = H|1\rangle = \frac{x+z}{\sqrt{2}}$

$YXY = -X$	$XXX = X$	$HXH = Z$	$SXS^\dagger = Y$
$XYX = -Y$		$HYH = -Y$	$SYS^\dagger = -X$
$XZX = -Z$		$HZH = X$	$SZS^\dagger = Z$

Unitary Conjugation $A \rightarrow UAU^\dagger$ $UU^\dagger = I = U^\dagger U$

